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Performance Analysis of Stocks using Deep Learning Models

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Abstract

This paper explores into stock market analysis using computer models to predict stock price changes, particularly focusing on the AAPL dataset from Apple Inc. Accurate stock forecasts are vital for informed investment decisions. By comparing different models—Multilayer Perceptron (MLP), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU)—the study offers a comprehensive performance analysis of deep learning approaches. The use of different deep learning architectures allowed us to capture different aspects of the stock price data and assess their effectiveness in making predictions. The project's findings would enable us to make informed decisions about which model to choose for future stock price prediction tasks, benefiting traders, investors, and other stakeholders in the financial domain. LSTM exhibited the best predictive capability, capturing intricate patterns due to its ability to handle long-term dependencies GRU also performed well. However, the MLP model didn't do as well because it couldn't grasp the complex patterns in the data as effectively as the other two models. So, for forecasting stock prices, the LSTM model proved to be the most accurate and reliable choice. These findings offer valuable insights for financial practitioners aiming to enhance stock price prediction using deep learning techniques, aiding in smarter investment strategies.

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Keywords: Stock Analysis; Multilayer Perceptron; Recurrent Neural Network; Long short-term memory; Gated Recurrent Unit

1. Introduction

The stock market acts as a dynamic arena where individuals actively participate in buying and selling stocks, resulting in continuous price fluctuations. The main aim of stock investment is to identify companies with growth potential, driving upward trends in stock prices [2]. Traditionally, people manually analyse charts to understand stock movements, which is a complex task. However, advancements in technology have made it easier to leverage large datasets from different sources. Machine learning techniques, including Artificial Neural Network, Deep Learning, and LSTM, have been employed to generate highly accurate predictions.

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While some experts argue that predicting stock prices is not random but intricate, it's based on analysing past market data to extract insights. Traders examine historical patterns to make informed assumptions about future stock prices. However, it's crucial to understand that predictions might not always be precise due to the impact of numerous variables on stock prices [1]. Hence, traders should consider a wide range of information and exercise caution while making trading decisions. Time series analysis reveals recurrent patterns, trends, and cycles within data. It helps identify market phases like bullish or bearish trends, aiding informed investment choices. This highlights the importance of time series analysis and forecasting in understanding stock behaviour and enhancing investment success. Various methods predict stock prices, broadly categorized as fundamental analysis, time series forecasting, and technical analysis. Fundamental analysis evaluates financial metrics like sales, earnings, for long-term forecasting. In contrast, technical analysis uses historical stock prices to predict future trends, often through techniques like moving averages for short-term predictions. Lastly, time series forecasting studies historical price patterns to predict future movements [4]. By examining how stock prices evolve, we gain insights to guide investment decisions. These methodologies collectively provide valuable tools for successful navigation of the stock market.

Forecasting and modelling time series data play significant roles in data analysis. It involves studying data that is collected over time and is commonly used in fields like econometrics and operations research. Time series analysis holds significant prominence in the realms of analytics and data science [2]. Currently, there are many tools for predicting stock prices, utilized by investment managers, companies, and individual investors. However, some of these tools can be expensive and not easily accessible for beginners or individuals.

2. Literature Review

In recent years, deep learning has emerged as a promising field of study, finding applications in tasks such as text classification, image processing, speech recognition, and more. It has also proven effective in time-series prediction due to its ability to handle chaotic data, randomness, and nonlinearity. Investment managers and brokers use various methods to understand stocks better. They look at how well a stock is doing compared to the overall market, company news, and what experts say about potential returns. All of this helps them predict how the stock might perform in the future, both in the short and long term.

In a study conducted by Nikou et al. (2019), they investigated various machine learning methods like Support Vector Machine (SVM), Random Forest, and Deep Learning. After comparing these methods, they found that the LSTM model, a special case of Recurrent Neural Network (RNN) within deep learning, demonstrated superior performance in predicting stock close prices compared to the other techniques [3].

Kavinnilaa J. et al. employed an ethically accurate forecasting technique to predict stock market indices. The LSTM model, surpassing other methods, excels due to its capacity to grasp long-term patterns in data. Advancing from RNN, it improves information management. Utilizing precise precision forecasting, the researchers offer valuable insights for investors, analysts, and individuals in making informed stock market decisions [5].

Chong et al. developed an advanced stock market prediction model using deep neural networks. Instead of relying on pre-existing knowledge, their model uses deep feature learning to identify essential patterns in stock return data. The model was tested on high-frequency data from the Korean stock market. This approach allows the system to extract abstract features from the information without needing prior knowledge of the predictors, making it a powerful tool for predicting stock market trends [6].

A. Objective

One of the most significant advantages of price prediction in the stock market is the potential to maximize profits while minimizing risks. By accurately forecasting stock prices, investors can make well-informed decisions on when to buy or sell securities, optimizing their investment returns and reducing potential losses. This helps investors strategically manage their portfolios and make informed choices about the timing of their transactions. In summary, stock market forecasting is a complex task due to the market's volatility and the multitude of factors influencing it. However, by leveraging advanced analytical techniques and financial indicators, accurate predictions can help investors navigate the market and make sound investment decisions to enhance profitability and mitigate risks.

3. Methodologies

In this paper, our aim is to leverage advanced Artificial Neural Network (ANN) and Recurrent Neural Network (RNN) models, namely Multilayer Perceptron (MLP), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) models, to analyse, pre-process, and predict stock prices accurately. The flowchart outlines the stock price forecasting process using various deep learning models: LSTM, MLP, and GRU. It starts with collecting historical stock price data as the input dataset. Flow chart of the proposed system is shown in figure 1.

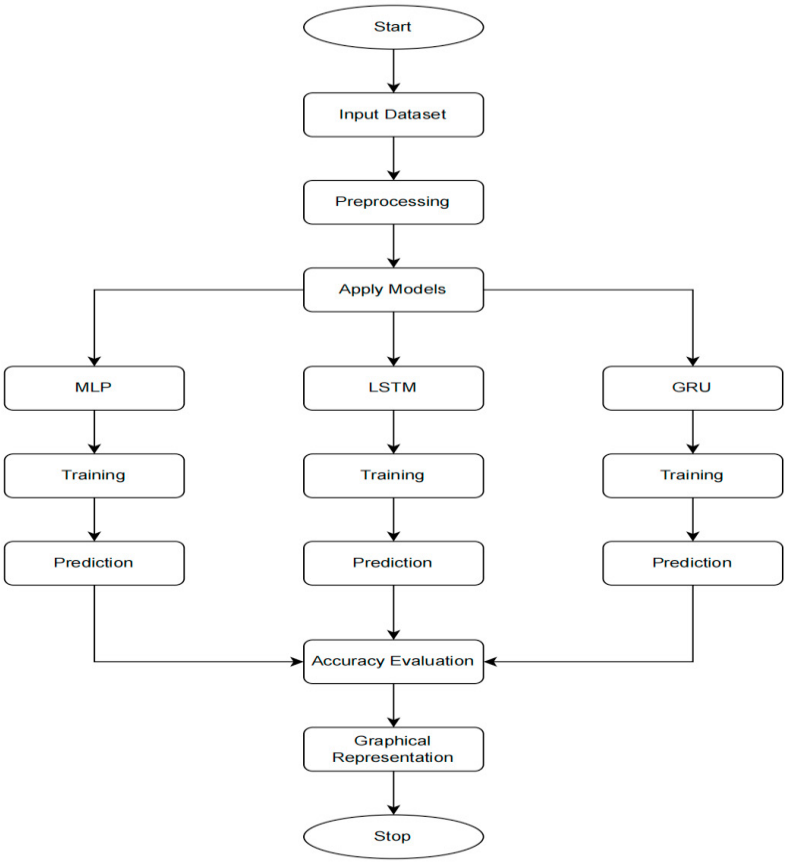


Fig 1- Flow chart of proposed system

3.1. Dataset Collection

We collect the AAPL (Apple Inc.) stock price dataset to use for our stock price forecasting analysis. The dataset is from July-15-2005 to May-15-2023, which will be obtained from the Yahoo Finance website. To ensure efficient data handling, store the fetched data in a pandas Data Frame, which will make it easier to manipulate and analyse. The dataset encompasses 4489 rows and 7 columns. Sample Dataset is shown in figure 2.

	Date	Open	High	Low	Close	Adj Close	Volume
0	2005-07-15	1.463214	1.484643	1.445000	1.483929	1.261255	687682800
1	2005-07-18	1.478929	1.503571	1.477500	1.481786	1.259433	586297600
2	2005-07-19	1.482857	1.543929	1.466786	1.542500	1.311036	671062000
3	2005-07-20	1.530714	1.564286	1.523214	1.558214	1.324393	453395600
4	2005-07-21	1.560714	1.572857	1.532143	1.546071	1.314072	404264000

Fig 2- Sample Dataset

- Open - Open price of a stock is the initial price at which it starts trading at the beginning of a trading day.
- High - The high price refers to the maximum price at which the stock was traded throughout a trading day.
- Low - The low price is the minimum price at which the stock was traded throughout a trading day.
- Close - The Close Price is the last price at which a stock is traded on a particular day.
- Volume - Volume is measured as the total number of shares bought and sold during a given trading period.

3.2. Data Pre-processing

In data pre-processing after data collection the very first step is addressing missing values, to check for missing values we've used pandas functions 'sum' and '.isnull()' and found that there is no any missing values in our dataset. Subsequently convert 'Date' column into Date Time format, which making it easier to handle time-based indexing and manipulation. After that set the 'Date' column as the index of the data frame for time-series analysis, which enabling us to explore the data over different time periods. We used the 'High' column from the dataset to train our model this column shows the highest share value, which helps us understand the stock's best performance. Split the dataset into 85-15 ratio for the train-test split. This allocation ensures that 85% of the data is dedicated to training the model and remaining 15% of the data is reserved for testing.

We are using RNN models LSTM and GRU, so it is important to convert the data into sequences with a specific timestep to capture temporal dependencies effectively. We set timestep to 60, since each sequence will now contain data from the past 60 days. Apply MinMaxScaler for consistent feature scaling, which ensures that the input features are on a similar scale.

3.3. MLP

The Multi-Layer Perceptron (MLP) stands as a widely used artificial neural network in machine learning, serving tasks such as classification and regression. Its name is derived from the multiple interconnected layers of nodes, also called artificial neurons [7]. In this paper, we inspect the crucial elements of MLP, encompassing its architecture, training procedures, and diverse applications across fields. Understanding its inner workings offers valuable insights into information processing through layers, enabling effective learning of intricate patterns in data[9]. The adaptability and versatility of MLP make it a fundamental and powerful tool for addressing diverse problems in different industries, leading to notable advancements in various domains.

MLP encounter difficulties as image pattern classifiers due to their limitations in capturing spatial relationships, sensitivity to variations, and inefficiencies in feature extraction. When used for images, they tend to flatten input, making them overly responsive to minor changes and difficult to local pattern recognition. MLP Architecture is shown in figure 3.

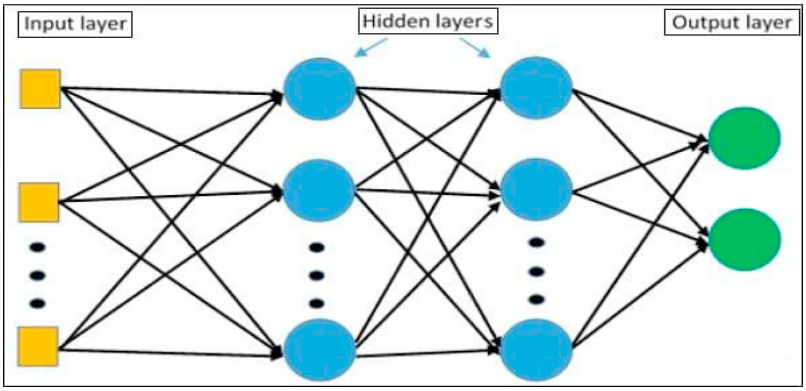


Fig 3- MLP Architecture

In multilayer neural networks should not use linear functions as activation functions because using linear function limits the network to representing linear relationships, hindering its ability to understand and learn complex and nonlinear patterns in data. To address this limitation, non-linear activation functions, such as ReLU or sigmoid, are commonly utilized to enable the network to learn.

3.4. LSTM

LSTM (Long Short-Term Memory) is a specialised variety of neural network used for handling sequences of data, like sentences or time series. It's an improved version of the traditional RNN (Recurrent Neural Network). LSTM was designed to solve the problem of vanishing gradients in traditional neural networks. This issue made it hard for the network to learn from long sequences of data and remembering long-term information [3].

LSTM networks are explicitly designed to address the long-term dependency problem that standard recurrent neural networks RNNs face. LSTMs overcome this problem through the incorporation of specialized memory cells and gating mechanisms. The key components of an LSTM include:

- Memory cell: It acts like a conveyor belt running through the whole sequence. This cell can store important information and carry it across different steps in the sequence [10-16].
- Input Gate (i): This gate determines the extent to which new information is incorporated into the cell state. It evaluates both the present input and the past memory and decides what's essential to keep.
- Forget Gate (f): This gate chooses which data from the cell state to discard. It also looks at the current input and past memory to figure out what's no longer important.
- Output Gate (o): The output gate chooses what information to give as a result from the cell state. It combines the current input and past memory to produce the output for the current step.

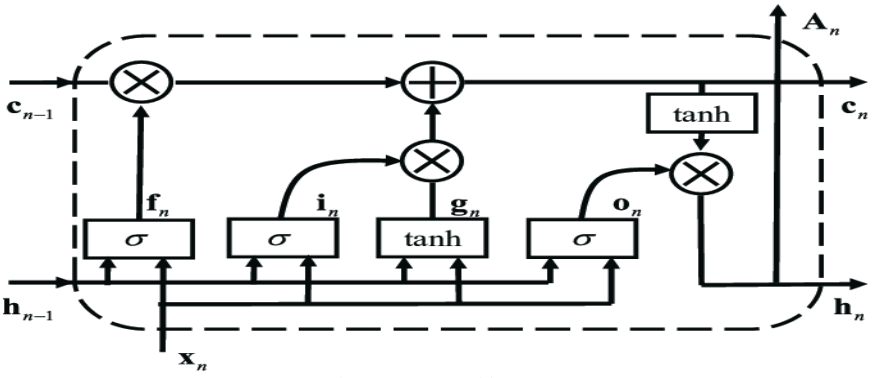


Fig 4- LSTM Architecture

LSTM models are essential for stock price prediction because stock price data is inherently sequential, with each data point dependent on its preceding ones. LSTMs, designed for sequential data, can capture patterns and trends over time, making them well-suited for modeling stock price movements. LSTM Architecture is shown in figure 4.

3.5. GRU

The Gated Recurrent Unit (GRU) shares a similar structure with the LSTM model, albeit with a simpler memory cell design. GRU streamlines its memory cell by incorporating just two gates: the reset gate and the update gate. The reset gate determines which data elements to overlook when processing new data, while the update gate manages the ongoing adjustment of the memory cell's state [8]. GRU stands out due to its quicker training time and good performance with limited data. GRU Architecture is shown in figure 5.

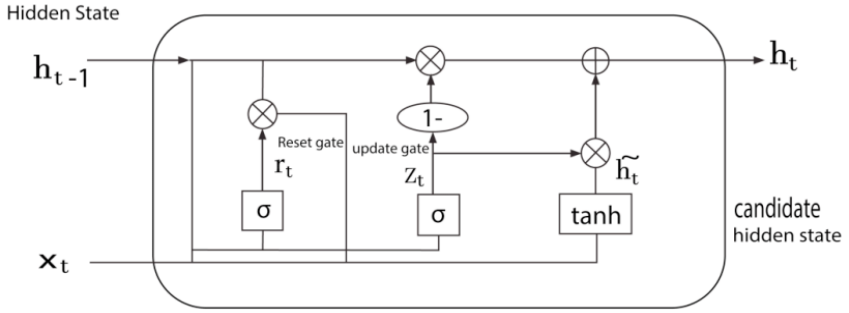


Fig 5- GRU Architecture

4. Apply Models and Evaluate the Results

After data preprocessing the refined dataset is utilized to apply MLP, LSTM, and GRU models on our training dataset and evaluate the performance of models using several evaluation metrics with different number of epochs using 10, 20, 30, 40, and 50 epochs respectively. These evaluation metrics are as follows-

- Mean Squared Error (MSE): MSE computes the average of squared difference between actual and predicted values.
- Root Mean Squared Error (RMSE): RMSE is the square root of MSE, denoting error in the target variable's units.
- Mean Absolute Error (MAE): It calculates the mean of absolute differences between observed and predicted values.
- R-squared**:** It serves as the coefficient of determination, representing the proportion of predictable variance in the dependent variable based on independent variables.

4.1. Applying MLP model and Evaluating Results

Initiating the MLP regression model with 100 hidden layers, we apply it to the training dataset. Following training, the model undergoes testing on test dataset, and performance evaluation metrics such as MSE, RMSE, and MAE are computed, presenting results in Table 1 with increasing number of epochs. R-squared is used to express the model's accuracy percentage. Additionally, Figure 6 depicts a graphical representation correlating date and prices, provide visual insight into the variance between predicted and observed values, which gives a comprehensive understanding of the model's predictive capabilities.

Table 1: Results of MLP

No. of Epochs	MAE	MSE	RMSE	Accuracy
10	11.85	184.18	13.57	40%
20	9.67	139.72	11.82	54%

30	8.63	113.73	10.66	63%
40	8.22	105.62	10.27	65%
50	7.01	81.66	9.03	73%

In summary, increasing the number of epochs allows the model to learn and improve its forecasting performance, but careful evaluation and validation are necessary to prevent overfitting and ensure accurate predictions on unseen data.

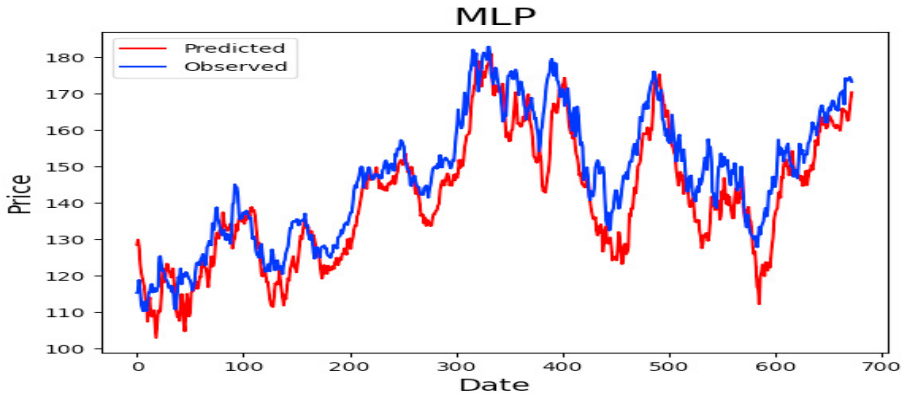


Fig 6- Graphical Representation of MLP Results

4.2. Applying LSTM model and Evaluating Results

Implementing a Long Short-Term Memory (LSTM) neural network for regression using Keras. Comprising four LSTM layers, each with 50 units, and interspersed with dropout layers (with dropout rate 0.2) to mitigate overfitting, the model captures temporal dependencies in sequential data. A dense output layer yields regression results. Compiled with 'rmsprop' as the optimizer and Mean Squared Error as the loss function, the model undergoes training on the dataset for 50 epochs with a batch size of 32. After implement these steps LSTM model is trained on the training data. After training, the model undergoes testing, and performance evaluation metrics such as MSE, RMSE, and MAE are computed, then presenting results in Table 2.

Table 2: Results of LSTM

No. of Epochs	**MAE**	**MSE**	**RMSE**	**Accuracy**
10	6.59	66.07	8.13	71%
20	6.31	60.05	7.75	74%
30	5.74	50.04	7.07	78%
40	5.54	47.01	6.86	79%
50	5.03	37.96	6.16	83%

During this analysis of LSTM model, we examine how the number of epochs impacts the model's error (loss) during training. As the epochs increase, the model iterates over the training data, fine-tuning its parameters to minimize the difference between predicted and actual stock prices. With more epochs, the model gains better insights into the main patterns in the data; it leads to a decrease in error. This is a common behaviour during the learning process, where the model's accuracy improves with more exposure to the data. In figure 6 represents the relation between time and prices, the graph visually reveals how well the model predicts, highlighting the variances

between predicted and actual values. Graphical Representation of LSTM Results is shown in figure 7.

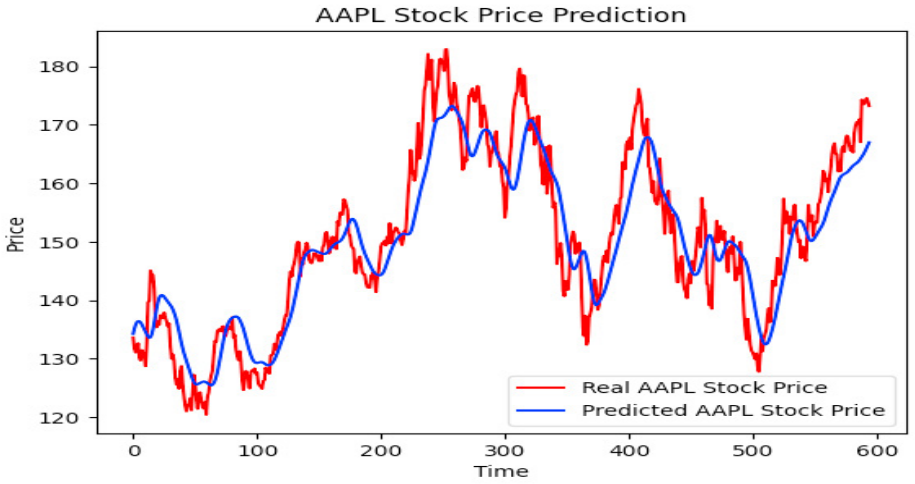


Fig 7- Graphical Representation of LSTM Results

4.3. Applying GRU model and Evaluating Results

Constructing a Gated Recurrent Unit (GRU) neural network for regression using Keras, four GRU layers with 50 units each and hyperbolic tangent (tanh) as an activation function are sequentially added, each followed by a dropout layer (dropout rate of 0.2) to prevent overfitting. A dense output layer with one unit produces regression output and the model is compiled with Stochastic Gradient Descent (SGD), specifying parameters like learning rate (0.01) and momentum (0.9), and using Mean Squared Error as the loss function. Training occurs over 10 to 50 epochs with a batch size of 150. This architecture aims to capture temporal dependencies in sequential data and is subject to hyper parameter tuning for optimal performance. Table 3 showcases improved forecasting accuracy with an extended number of epochs, indicating a positive correlation between training duration and performance.

Table 3: Results of GRU

No. of Epochs	MAE	MSE	RMSE	Accuracy
10	7.14	73.25	8.56	69%
20	6.87	60.05	8.33	70%
30	6.09	55.13	7.43	75%
40	5.98	54.03	7.35	77%
50	5.75	49.89	7.66	79%

When we implement and analyse the GRU model, we found how the number of epochs impacts the model's error (loss) during training. As the epochs increase, the model iterates over the training data, fine-tuning its parameters to minimize the difference between predicted and actual stock prices. Figure 6 depicts a graphical representation of the

relationship between time and prices, highlighting the disparity between predicted and observed values. The graph provides a visual insight into the model's predictive performance, emphasizing the variation in price predictions over different time period.

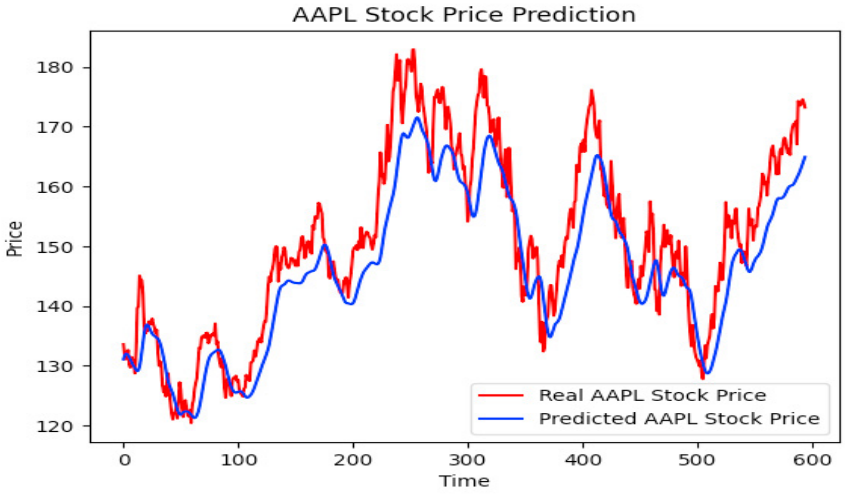


Fig 8- Graphical Representation of GRU Results

5. Conclusion

In this paper, we comprehensively compared three popular deep learning models: Multilayer Perceptron (MLP), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU), for stock price prediction using AAPL's historical data. Models were trained, evaluated, and tested for accuracy in forecasting. The LSTM model excelled due to its grasp of long-term and complex stock price patterns. GRU effectively captured time-related patterns. Conversely, MLP struggled with complex patterns. The study concludes that for accurate and reliable stock price prediction, LSTM is the preferred choice, benefiting financial stakeholders by informing better decision-making.

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